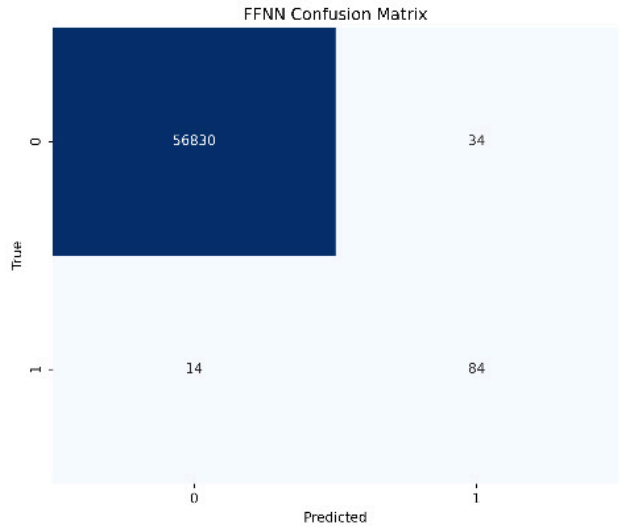
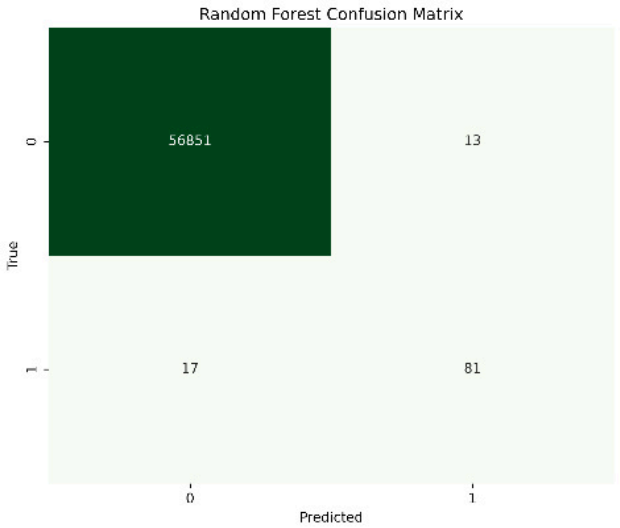
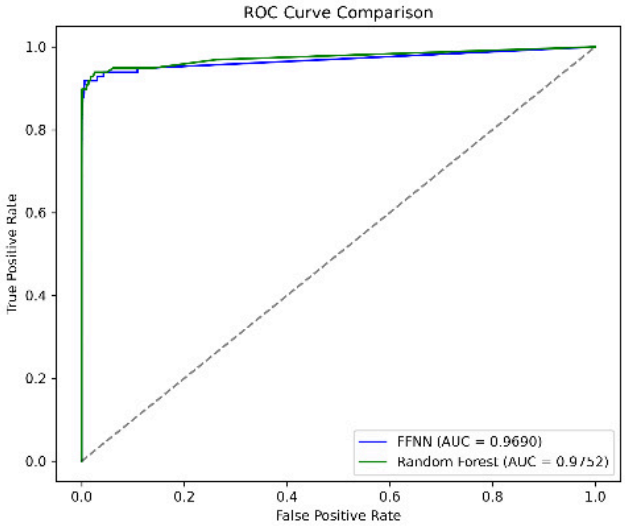
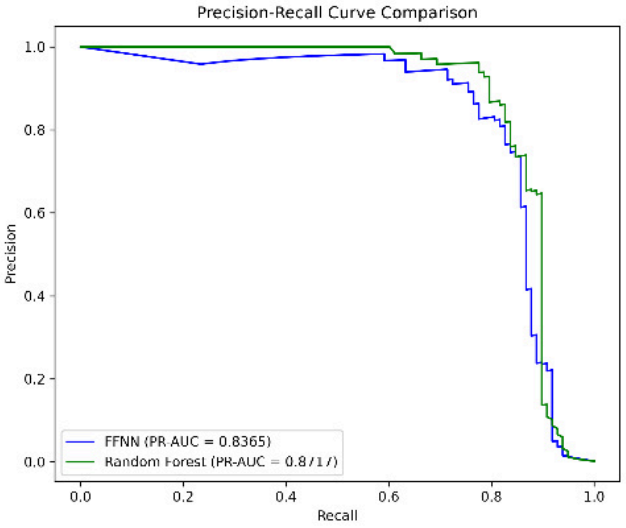
Part D

1. <u>Which model is best between Model 1 and Model 2?</u>
   It is clear that the Random Forest model was overall the better model at predicting fraud amongst the test set.










```
========================================
--- MODEL COMPARISON SUMMARY ---
========================================
  Metric    FFNN  Random Forest
F1-Score 0.7778          0.8438
 ROC-AUC 0.9690          0.9752
  PR-AUC 0.8365          0.8717
========================================
```

We can see that the Random forest had a significantly lower amount of False Positives meaning that it raised a false alarm at a much lower rate. If we keep our goal in mind, which was to reduce the amount of False positives, then the random forest model was successful. However, on the other hand, the Random Forest called more false positives than the Neural Network. In the context of fraud detection, the cost of a false positive is extremely high. This means that the model missed multiple cases of actual fraud occurring which could lead to great repercussions for the client's financial well-being and the business itself. Ultimately this could be amended by optimizing the random forests hyperparameters so that it does not miss any occurrences of fraud.

Overall, the random forest model performed better with a higher F-1 Score, a slightly higher ROC-AUC and a higher PR-AUC score. The trade-offs for model precision and recall are worth the slightly higher number of actual frauds missed because we can tune the model to catch them with our hyperparameters.

Ultimately the random forest was the better model to solve the business problem of less false positive flags. This will reduce customer friction leading to less queries and more operational capacity to solve other meaningful queries for clients.

2. <u>How can the selected model be improved?</u>
   Firstly, we can optimize the hyperparameters. The current hyperparameters were chosen based on the grid search. We could try alternative methods for hyperparameter tuning such as Bayesian optimization, which may produce different results.

   We could also change the threshold which flags transactions as fraudulent. Amending the threshold could produce better results and ensure that we do not miss any fraudulent transactions.

   Unfortunately due to the nature of the project the actual data which we had was confidential. If we had actual feature names we would be able to create a more robust model which takes into account individual client behaviour, clusters it based on past transactions and allows us to evaluate if future transactions fall outside of their normal behaviour. The model would greatly benefit from greater context in data, which would allow us to create multiple useful features allowing for better inference.

3. <u>How will the selected model need to be adapted to work with your requested data from SS1?</u>
   I am currently aware of how to handle the data that would be provided. I know that there is great class imbalance and this will be addressed with SMOTE. Our whole dataset needs to be scaled, in order to accurately draw insights from the data. With the expected dataset, I will have even more context regarding the data as it most likely will be supplied with feature names. The Dataset used experimentally was credit card data, which is not exactly retail investment data, however it does allow us to apply the correct set of principles to allow for accurate inference.

I have accurately mapped out the pipeline for how to handle such fraud detection data and I simply need to prepare the data and input it to achieve results.

4. Do your results align with the related literature from Part A?
   Yes. Many of the publications mentioned that fraud data is heavily imbalanced due to the rare nature of fraud. To combat this most publications suggested using either over-sampling or undersampling methods to balance out the data. However they stated that undersampling was sub-optimal as it was also possible to reduce the total number of observations in our datasets. Fraud data is very difficult to obtain so it is of utmost importance to preserve as much data as possible. So because of this I used SMOTE which led to good results in terms of recall and precision with the chosen model (Random Forest)

   My train test split for my neural network was 20% Test / 16% Validation / 64 Train. This deviates from the publications which mentioned 70/15/15 - train/validation/test splits. My split was similar however I believe that my Neural network could have benefited from having a greater amount of training data.

   The literature has stated that tree based models handle skewed data well as they are good at inference when outliers are involved. This is true for our project as after training the random forest model handles the outliers (frauds) well and in a balanced manner.